

NATIONAL LEVEL HACKATHON

PHASE 1 – IDEA SUBMISSION

Domain: EdTech & Skill Development

Project Title: SkillBridge – Smart Learning & Career Skill Development Platform

A student-centric digital platform that connects learning, skill assessment, practical projects and career-oriented guidance in one ecosystem. The goal is to help students identify skill gaps, follow personalized learning paths and build demonstrable, industry-relevant skills.

1. Problem Understanding

Students today have access to a huge amount of online educational content, but access to information does not always translate into employable skills. Learners often face difficulty choosing the right courses, understanding what skills are required for a career, measuring their actual proficiency and getting opportunities to apply what they learn.

Major challenges

- Fragmented learning resources across multiple platforms.
- Lack of personalized learning paths based on a student's current level and career goal.
- Limited practical exposure through projects and real-world challenges.
- Difficulty identifying skill gaps and tracking improvement.
- Weak connection between academic learning, industry expectations and career preparation.
- Students from smaller institutions may have limited access to structured mentorship and guidance.

Problem Statement

How can we create an affordable and scalable platform that converts learning into measurable, practical and career-relevant skills for students?

2. Proposed Solution

SkillBridge is proposed as an integrated EdTech and skill-development platform. A student creates a profile, selects a career interest, completes a baseline skill assessment and receives a personalized learning roadmap. The platform then combines learning modules with quizzes, practical projects, progress tracking and career guidance.

Core approach

- Assess → identify current knowledge and skill gaps.
- Recommend → generate a personalized learning roadmap.
- Learn → provide structured concepts, resources and practice.
- Apply → assign practical projects and challenges.
- Measure → track assessments, projects and skill progress.
- Prepare → connect acquired skills with career roles and portfolio building.

3. System Architecture

The system follows a modular architecture so that learning, assessment, recommendation and career services can work together while remaining scalable.

Layer	Components	Purpose
1. User Layer	Student / Mentor / Admin	Secure access and role-based interaction.
2. Presentation Layer	Web/Mobile UI, Dashboard	Courses, assessments, projects and progress visualization.
3. Application Layer	Learning, Assessment, Project, Career modules	Business logic and platform workflows.
4. Intelligence Layer	Recommendation + AI/NLP services	Personalized learning and skill-gap recommendations.
5. Data Layer	User DB, Course DB, Assessment DB, Project DB	Stores profiles, progress, content and results.
6. External Services	Cloud, notifications, optional job/internship APIs	Deployment, communication and future integrations.

Architecture Flow

Student	→	Web/Mobile UI	→	Backend APIs	→	AI/Recommendation	→	Database
		Courses • Tests • Projects		Auth • Logic • Progress		Skill Gap • Roadmap		Profiles • Results

4. Key Features

Personalized Learning Roadmap: Creates a learning path based on interests, baseline assessment and target career.

Skill Assessment Engine: Uses quizzes and assessments to estimate proficiency and highlight learning gaps.

Project-Based Learning: Recommends practical projects that allow students to demonstrate their skills.

Progress Dashboard: Shows course completion, assessment scores, projects and skill growth.

AI Learning Assistant: Provides concept explanations, practice questions and learning guidance.

Career Skill Mapping: Maps learning outcomes to relevant job roles and identifies missing skills.

Mentor & Peer Support: Enables feedback, collaboration and structured guidance.

Digital Skill Portfolio: Collects projects, achievements, certificates and assessment results in one profile.

5. Technology Stack

Component	Technology	Role
Frontend	React / HTML / CSS / JavaScript	Responsive student dashboard and user interface
Backend	Python FastAPI / Node.js	APIs, authentication and business logic
Database	MySQL / PostgreSQL / Firebase	User, course, assessment and progress data
AI / ML	Python, scikit-learn / NLP models	Recommendation and skill-gap analysis
Cloud	AWS / Azure / Firebase	Scalable hosting and deployment
Security	JWT, HTTPS, role-based access	Secure authentication and data access
Version Control	Git + GitHub	Team collaboration and code management

6. Implementation Plan

The solution will be developed in iterative phases so that a functional Minimum Viable Product (MVP) can be demonstrated during the hackathon.

Phase	Activities	Deliverable
Phase 1 – Research	User needs, problem validation, feature prioritization	Requirements + user journey
Phase 2 – UI/UX	Design student dashboard, assessment and learning screens	Clickable prototype
Phase 3 – Backend	Database, authentication, APIs and core modules	Working backend
Phase 4 – Intelligence	Skill-gap logic and personalized recommendations	Recommendation module
Phase 5 – Integration	Connect frontend, backend, database and AI module	Functional MVP
Phase 6 – Testing	Functional, usability and security testing	Tested prototype
Phase 7 – Deployment	Cloud deployment, demo preparation and documentation	Hackathon-ready demo

7. MVP Demonstration Flow

- Student registers and selects a target career/skill area.
- Student completes a short baseline assessment.
- System identifies strengths and skill gaps.
- Personalized roadmap is generated.
- Student completes learning content and quizzes.
- System recommends a practical project.
- Progress dashboard updates and the project is added to the skill portfolio.

8. Expected Impact

SkillBridge is intended to make learning more structured, practical and outcome-oriented. It can help students understand what to learn, why they need to learn it, how to practice it and how to demonstrate their skills. The platform can also help institutions monitor student skill development and encourage project-based learning.

9. Innovation & Competitive Advantage

- **Learning-to-Skill Pipeline:** connects learning resources directly with assessments and practical projects.
- **Personalization:** recommendations are based on the learner's current level and target career rather than a one-size-fits-all course list.
- **Evidence-Based Progress:** projects and assessments create measurable evidence of skill development.
- **Career Alignment:** learning goals are connected to skills expected for relevant career roles.
- **Scalable Model:** the platform can support multiple branches, domains, institutions and regional languages.

10. Feasibility & Scalability

The proposed solution uses widely available web technologies, cloud infrastructure and modular software components. A basic MVP can be developed with a limited set of courses, assessments and projects, then expanded progressively. The same architecture can support engineering, commerce, science, vocational and other learning domains.

11. Future Scope

- Integration with verified internships, competitions and job opportunities.
- Industry-created challenges and skill assessments.
- AI-assisted resume and portfolio generation.
- Multilingual content and regional-language learning.
- Institutional analytics for faculty and placement teams.
- Gamification through badges, levels and learning streaks.
- Integration with external learning platforms through APIs.

12. Conclusion

SkillBridge addresses the gap between education and employability by bringing personalized learning, skill assessment, practical projects, progress tracking and career guidance into one platform. The solution is designed to be student-centric, technically feasible and scalable. By shifting the focus from simply consuming educational content to building measurable skills, SkillBridge can help learners become more confident and career-ready.

Phase 1 Submission – National Level Hackathon

Domain: EdTech & Skill Development

Submission fields to customize before uploading: Team Name • Team Members • College/Institution • Team Leader • Contact Details • Mentor • GitHub/Prototype Link.